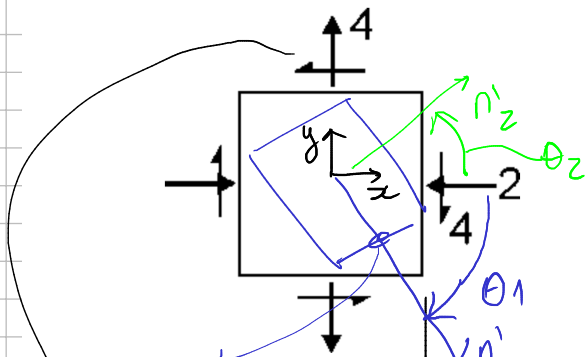


Q104 - E1



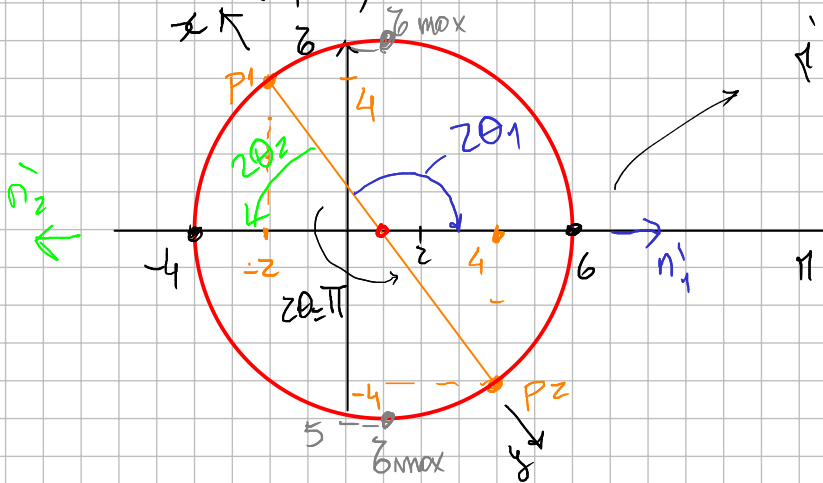
$$\underline{\underline{\tau}} = \begin{bmatrix} -2 & -4 \\ -4 & 4 \end{bmatrix}$$

$\sigma = 6$
 $\tau = 0$

$P_1 = (-2, 4)$
 $P_2 = (4, -4)$

Principal stresses:

$$\underline{\underline{\sigma}} = \begin{bmatrix} 6 & 0 \\ 0 & -4 \end{bmatrix}$$



$$\sigma_{\max} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{\min}$$

$$\underline{\underline{\tau}} = \begin{bmatrix} -2 & -4 \\ -4 & 4 \end{bmatrix}$$

$$\sigma_{\max/\min} = \frac{-2 + 4}{2} \pm \sqrt{\left(\frac{-2 - 4}{2}\right)^2 + (-4)^2} = 1 \pm 5 \rightarrow \sigma_{\max} = 6$$

$$\rightarrow \sigma_{\min} = -4$$

$$\tan 2\theta = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{2 \cdot -4}{-2 - 4} = \frac{8}{6} = \frac{4}{3} \Rightarrow \theta_2 = 26,56^\circ$$

$$\theta_1 = 90^\circ + 26,56^\circ = 116,56^\circ \rightarrow$$

$$= 26,56^\circ - 90^\circ = -63,44^\circ$$

$$\tau_{\max} = \frac{\sigma_{\max} - \sigma_{\min}}{2} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

$$\sigma_{\max} = \frac{6 - (-4)}{2} = 5$$